04. Database design: tables and indexes

Status	Draft v1.1. The complete list of tables and indexes. Design rules, migration, Redis, personal data and the MVP delta are in `04b-database-notes.md`
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1. Overview

- **Two PostgreSQL databases** on one server: `shazlo_core` (35 tables, used by the Core service) and `shazlo_vendor` (23 tables, used by the Vendor-admin service). 58 tables in total.
- Every table has `id` (uuid) plus `created_at` and `updated_at`, except tables marked **bigint id** (append-only event tables) and tables whose primary key is named in the Columns cell.
- Money is stored in paise (`bigint`). Times are `timestamptz` (UTC).
- Foreign keys exist only inside one database. A column marked "(plain id)" points into the other database and has no foreign key.

2. Core database: `shazlo_core` (35 tables)

#	Table	Columns	Indexes and constraints
1	`users`	email, phone, password, first_name, last_name, display_name, is_guest, is_active, is_staff, is_superuser, last_login, deleted_at	Unique lower(email) where not null; unique phone where not null; (is_guest, created_at)
2	`user_profiles`	**user_id (PK, FK users)**, gender, date_of_birth, language, marketing_opt_in, rewards_balance, total_swipes, total_likes	PK user_id
3	`addresses`	user_id (FK), recipient_name, phone, line1, line2, landmark, city, state, pincode, is_default	(user_id); unique (user_id) where is_default
4	`devices`	user_id (FK), platform, push_token, app_version, last_seen_at	Unique push_token; (user_id)
5	`social_identities`	user_id (FK), provider, subject, email_at_link	Unique (provider, subject)
6	`consents`	user_id (FK), kind, version, accepted_at	(user_id, kind)
7	`refresh_tokens`	family_id, user_id (FK), token_hash, expires_at, revoked_at, replaced_by	Unique token_hash; (family_id)
8	`catalog_items`	source, vendor_id (plain id), vendor_name, commission_bps, brand_name, title, description, category_slug, category_name, gender, fabric, attributes, min_price_paise, max_price_paise, images, variants, returnable, return_window_days, external_url, status, feed_eligible, version, source_updated_at	(category_slug, gender) where feed_eligible; (vendor_id); (status)
9	`product_embeddings`	**product_id (PK, FK catalog_items)**, model_version, embedding vector(512), source_image_url	PK product_id; **HNSW on embedding (cosine)**

#	Table	Columns	Indexes and constraints
10	`swipe_events` (bigint id)	user_id (FK), product_id (FK catalog_items), action, dwell_ms, session_id, slate_id, position, source, client_ts, idempotency_key	(user_id, id desc); (product_id, action); unique idempotency_key where not null
11	`user_taste_clusters`	user_id (FK), cluster_index, centroid vector(512), weight, updates_count	Unique (user_id, cluster_index)
12	`feed_impressions` (bigint id)	user_id (FK), product_id (FK), slate_id, position, served_at	(user_id, id desc)
13	`reward_events` (bigint id)	user_id (FK), kind, points, reason	(user_id, id desc)
14	`closets`	name, created_by (FK users)	(created_by)
15	`closet_members`	closet_id (FK), user_id (FK), role, joined_at	Unique (closet_id, user_id); (user_id)
16	`closet_items`	closet_id (FK), product_id (FK catalog_items), added_by (FK users), added_at	Unique (closet_id, product_id)
17	`closet_invites`	closet_id (FK), token, created_by (FK), expires_at, max_uses, uses, revoked_at	Unique token; (closet_id)
18	`carts`	user_id (FK)	Unique user_id
19	`cart_items`	cart_id (FK), variant_id (plain id), product_id (FK catalog_items), quantity	Unique (cart_id, variant_id); check quantity between 1 and 20
20	`inventory`	**variant_id (PK, plain id)**, product_id (FK catalog_items), vendor_id (plain id), on_hand, reserved, version	PK variant_id; (product_id); (vendor_id); check on_hand >= 0, reserved >= 0, reserved <= on_hand
21	`stock_reservations`	order_id (FK orders), variant_id, quantity, status, expires_at	(order_id); (expires_at) where status = 'held'
22	`orders`	number, customer_id (FK users), vendor_id (plain id), checkout_group_id, status, payment_method, payment_status, subtotal_paise, delivery_fee_paise, discount_paise, total_paise, commission_bps, commission_paise, vendor_net_paise, price_versions, customer_name, customer_phone, address_snapshot, note, placed_at, payment_due_at, confirmation_due_at, reservation_expires_at, confirmed_at, shipped_at, delivered_at, cancelled_at, cancel_reason, idempotency_key, version	Unique number; unique (customer_id, idempotency_key); (vendor_id, status, placed_at); (customer_id, placed_at desc); (checkout_group_id); (status, confirmation_due_at) where status = 'placed'; (payment_due_at) where payment_status = 'pending' and payment_method = 'online'
23	`order_items`	order_id (FK), variant_id, product_id, title, sku, size, color, image_url, unit_price_paise, quantity, line_total_paise	(order_id)
24	`order_status_events` (bigint id)	order_id (FK), from_status, to_status, actor_type, actor_id, reason	(order_id, id)
25	`payments`	order_id (FK), method, provider, provider_order_id, provider_payment_id, amount_paise, status, captured_at, raw	Unique (provider_payment_id, order_id) where provider_payment_id is not null; (order_id); (provider_order_id)
26	`refunds`	payment_id (FK), order_id (FK), amount_paise, status, reason, provider_refund_id, processed_at	(order_id); (payment_id)
27	`deliveries`	order_id (FK), courier_name, tracking_ref, status, fee_paise	Unique order_id

#	Table	Columns	Indexes and constraints
28	`delivery_events` (bigint id)	delivery_id (FK), status, note, payload	(delivery_id, id)
29	`returns`	order_id (FK), status, reason, requested_by, decline_reason, resolved_at	(order_id)
30	`return_items`	return_id (FK), order_item_id (FK), quantity	(return_id)
31	`outbox_events` (bigint id)	event_id, type, aggregate_id, version, payload, status, attempts, next_attempt_at, sent_at	Unique event_id; (status, next_attempt_at) where status = 'pending'
32	`inbox_events`	**event_id (PK)**, type, aggregate_id, version, received_at, processed_at	PK event_id; (aggregate_id)
33	`idempotency_keys`	scope, key, request_hash, response	Unique (scope, key)
34	`notification_log` (bigint id)	user_id (FK), order_id (FK), channel, template, status, provider_message_id, error	(order_id, id); (user_id, id)
35	`audit_log` (bigint id)	actor_id, actor_type, action, entity_type, entity_id, before, after, ip, user_agent	(entity_type, entity_id, id); (actor_id, id)

3. Vendor-admin database: `shazlo_vendor` (23 tables)

#	Table	Columns	Indexes and constraints
1	`vendors`	name, slug, status, owner_name, phone, email, city, state, instagram_or_website, about, logo_key, declared_price_min_paise, declared_price_max_paise, product_types, source, commission_bps, agreement_signed_at, agreement_ref, gstin, reviewed_by, reviewed_at, rejection_reason	Unique slug; unique lower(email) where status in ('applied', 'under_review'); (status, created_at)
2	`commission_rates`	vendor_id (FK), commission_bps, effective_from, changed_by, reason	(vendor_id, effective_from desc)
3	`vendor_users`	vendor_id (FK), email, phone, password_hash, role, status, invited_at, accepted_at, last_login_at	Unique lower(email); unique phone where not null; (vendor_id); unique (vendor_id) where role = 'vendor_owner'
4	`admin_users`	email, name, password_hash, status, totp_secret, last_login_at	Unique lower(email)
5	`refresh_tokens`	family_id, subject_type, subject_id, token_hash, expires_at, revoked_at, replaced_by	Unique token_hash; (family_id)
6	`vendor_policies`	vendor_id (FK), returnable, return_window_days, return_shipping_paid_by, processing_hours, notes	Unique vendor_id
7	`vendor_locations`	vendor_id (FK), label, line1, line2, city, state, pincode, is_default	(vendor_id); unique (vendor_id) where is_default
8	`vendor_status_events` (bigint id)	vendor_id (FK), from_status, to_status, actor_id, reason	(vendor_id, id)
9	`categories`	parent_id (FK categories), name, slug, level	Unique slug; (parent_id)

#	Table	Columns	Indexes and constraints
10	`products`	vendor_id (FK), title, description, category_id (FK), gender, fabric, attributes, status, rejection_reason, returnable, return_window_days, weight_grams, min_price_paise, max_price_paise, version, published_at	(vendor_id, status); (status, created_at); (category_id)
11	`product_variants`	product_id (FK), vendor_id (FK), sku, size, color, price_paise, mrp_paise, is_active	Unique (product_id, size, color); unique (vendor_id, sku); check price_paise > 0, mrp_paise >= price_paise
12	`product_images`	product_id (FK), key, position, angle, width, height, alt_text	Unique (product_id, position)
13	`product_status_events` (bigint id)	product_id (FK), from_status, to_status, actor_type, actor_id, reason	(product_id, id)
14	`import_jobs`	vendor_id (FK), created_by (FK vendor_users), kind, status, file_key, total_rows, ok_rows, error_rows, errors, finished_at	(vendor_id, created_at)
15	`vendor_orders`	**order_id (PK, plain id)**, number, vendor_id (FK), customer_id (plain id), status, payment_method, payment_status, subtotal_paise, delivery_fee_paise, total_paise, commission_bps, commission_paise, vendor_net_paise, customer_name, customer_phone, address_snapshot, placed_at, confirmation_due_at, confirmed_at, shipped_at, delivered_at, cancelled_at, cancel_reason, version	PK order_id; unique number; (vendor_id, status, placed_at); (customer_id)
16	`vendor_order_items`	order_id (FK vendor_orders), product_id, variant_id, title, size, color, image_url, unit_price_paise, quantity	(order_id)
17	`vendor_payouts`	vendor_id (FK), period_start, period_end, orders_count, gross_paise, commission_paise, net_paise, status, paid_at, reference	Unique (vendor_id, period_start, period_end)
18	`payout_lines`	payout_id (FK), order_id (FK vendor_orders), gross_paise, commission_paise, net_paise	Unique order_id; (payout_id)
19	`outbox_events` (bigint id)	event_id, type, aggregate_id, version, payload, status, attempts, next_attempt_at, sent_at	Unique event_id; (status, next_attempt_at) where status = 'pending'
20	`inbox_events`	**event_id (PK)**, type, aggregate_id, version, received_at, processed_at	PK event_id; (aggregate_id)
21	`idempotency_keys`	scope, key, request_hash, response	Unique (scope, key)
22	`vendor_notification_log` (bigint id)	vendor_id (FK), order_id (plain id), channel, template, status, provider_message_id, error	(order_id, id); (vendor_id, id)
23	`audit_log` (bigint id)	actor_id, actor_type, action, entity_type, entity_id, before, after, ip, user_agent	(entity_type, entity_id, id); (actor_id, id)

4. Enumerations

Name	Values	Database
Gender (profile)	`women`, `men`, `other`, `unspecified`	core
Product gender	`women`, `men`, `unisex`	both
Catalog item source	`vendor`, `scraped`	core

Name	Values	Database
Catalog item status	`approved`, `hidden`	core
Vendor status	`applied`, `under_review`, `approved`, `rejected`, `suspended`	vendor
Vendor user role	`vendor_owner`, `vendor_staff`	vendor
Product status	`draft`, `pending_review`, `approved`, `rejected`, `archived`	vendor
Order status	`placed`, `confirmed`, `shipped`, `delivered`, `cancelled`, `return_requested`, `returned`, `refunded`	core, vendor (copy)
Payment method	`cod`, `upi_on_delivery`, `online`	core, vendor (copy)
Payment status	`pending`, `paid`, `failed`, `refunded`	core, vendor (copy)
Reservation status	`held`, `committed`, `released`	core
Swipe action	`like`, `dislike`, `save`, `cart`, `order`	core
Closet member role	`owner`, `editor`	core
Event type	`product.published`, `product.updated`, `product.unpublished`, `vendor.status_changed`, `vendor.commission_changed`, `order.placed`, `order.status_changed`, `payment.captured`, `payment.refunded`, `order.confirmation_overdue`, `customer.deleted`	both